EXPERIMENT 11: Create and Execute Your First Container Using Docker

1. Aim

To use Docker to download an image, create a container and execute commands inside the container.

2. Procedure

1. Visit the official Docker website and install Docker Desktop or Docker Engine for the operating system.
2. Start the Docker service or Docker Desktop.
3. Open a terminal or command prompt.
4. Verify the Docker installation using docker --version.
5. Check the Docker service using docker info.
6. Download the official Ubuntu image from Docker Hub using docker pull ubuntu.
7. Verify the downloaded image using docker images.
8. Create and start a container interactively using docker run -it --name mycontainer ubuntu bash.
9. Observe the container shell prompt and execute a command such as echo Hello Docker.
10. Exit the container using the exit command.
11. List all containers, including stopped containers, using docker ps -a.
12. Start the stopped container again using docker start -ai mycontainer.

3. Program

Docker commands:

```
docker --version
docker info
docker pull ubuntu
docker images
docker run -it --name mycontainer ubuntu bash

# Run inside the container:
echo "Hello Docker"
exit

# On the host:
docker ps -a
docker start -ai mycontainer
```

4. Output of Program

```
docker pull ubuntu
Status: Downloaded newer image for ubuntu:latest

$ docker run -it --name mycontainer ubuntu bash
root@container:/# echo "Hello Docker"
Hello Docker
root@container:/# exit
```

5. Result

Thus, Docker was used to download an Ubuntu image, create a container and execute commands inside it successfully.

EXPERIMENT 11: Create and Execute Your First Container Using Docker

1. Aim

To install and configure Docker, download an Ubuntu image from Docker Hub, create a container using the downloaded image and execute commands inside the container.

2. Procedure

1. Start the operating system and open a web browser. 2. Visit the official Docker website and download Docker Desktop or Docker Engine according to the operating system. 3. Install Docker by following the installation instructions. 4. Start the Docker service or Docker Desktop and wait until the Docker engine becomes ready. 5. Open a terminal or command prompt to execute Docker commands. 6. Verify the Docker installation using docker --version. 7. Execute docker info to check the Docker engine configuration and system information. 8. Download the official Ubuntu image from Docker Hub using docker pull ubuntu. 9. Wait until the image download process is completed. 10. Execute docker images to verify that the Ubuntu image is available locally. 11. Create and start a container using docker run -it --name mycontainer ubuntu bash. 12. Observe that an interactive shell is opened inside the Ubuntu container. 13. Execute a simple command such as echo "Hello Docker" inside the container. 14. Execute other basic Linux commands such as pwd, ls and whoami to verify the container environment. 15. Exit the container using the exit command. 16. Return to the host terminal and execute docker ps -a to display all containers, including stopped containers. 17. Identify the container named mycontainer in the displayed list. 18. Restart the stopped container using docker start -ai mycontainer. 19. Verify that the interactive shell is available again. 20. Exit the container and confirm that the container can be started and stopped successfully.

4. Output of Program

Docker Image Download: $ docker pull ubuntu Status: Downloaded newer image for ubuntu:latest Container Execution: $ docker run -it --name mycontainer ubuntu bash root@container:/# echo "Hello Docker" Hello Docker root@container:/# exit Container Verification: $ docker ps -a CONTAINER ID IMAGE STATUS xxxxxxxx ubuntu Exited Explanation: Docker images act as templates used to create containers. A container is an isolated execution environment created from an image. In this experiment, an Ubuntu image is downloaded and used to create an interactive container.

5. Result

Thus, Docker was successfully installed and configured. The Ubuntu image was downloaded from Docker Hub, a container was created and started, and basic commands were executed successfully inside the container.